Integral with Respect to a Measure

Definition

Let $(X, \mathcal{A}, \mu)$ be a measure space and let

$$f : X \to \mathbb{R}$$

be a measurable function. The integral

$$\int_X f \, d\mu$$

is defined in stages.

Indicator Functions

For a measurable set $A \in \mathcal{A}$, define

$$\int_X \mathbf{1}_A \, d\mu = \mu(A).$$

Nonnegative Simple Functions

If

$$s = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i},$$

where $a_i \geq 0$ and $A_i \in \mathcal{A}$, define

$$\int_X s \, d\mu = \sum_{i=1}^{n} a_i \mu(A_i).$$

Nonnegative Measurable Functions

If $f : X \to [0, \infty]$ is measurable, define

$$\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu : 0 \leq s \leq f, \ s \text{ simple} \right\}.$$

General Measurable Functions

For a real-valued measurable function f, write

$$f^+ = \max\{f, 0\}, \qquad f^- = \max\{-f, 0\}.$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

If at least one of $\int_X f^+ \, d\mu$ and $\int_X f^- \, d\mu$ is finite, define

$$\int_X f \, d\mu = \int_X f^+ \, d\mu - \int_X f^- \, d\mu.$$

If both are finite, then f is integrable and

$$\int_X |f| \, d\mu < \infty.$$

Interpretation

The notation does not necessarily mean integration with respect to Lebesgue measure. The measure is the object written after d.

For example:

$$\int_{\mathbb{R}^d} f \, d\mu$$

means integration with respect to μ. If μ is Lebesgue measure, this is the usual Lebesgue integral. If μ is a probability measure, it is the expectation of f under μ:

$$\int_{\mathbb{R}^d} f \, d\mu = \mathbb{E}_\mu[f].$$

Absolutely Continuous Case

If μ has density ρ with respect to Lebesgue measure, then

$$d\mu(x) = \rho(x) \, dx$$

and therefore

$$\int_{\mathbb{R}^d} f \, d\mu = \int_{\mathbb{R}^d} f(x)\rho(x) \, dx.$$